Kalman Filter Theory and Applications
Part III: State Space Concepts

https://github.com/musicarroll/kalman_course

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Part III

State Space Concepts

Topics

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- Understanding the Equations: Heuristic Introduction

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- Equation Drilldown: Taking the Equations Apart

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- **State Space Concepts** ($\geq$8 videos)

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State Space Concepts

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Dynamic Systems

A Note on Notation

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- The first n derivatives are the $0^{\text{th}}, 1^{\text{st}}, \ldots, (n-1)^{\text{st}}$ derivatives of the output function y
- These values capture the "state" of the dynamic system at any time after the time of the initial conditions

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- Taylor's theorem: If a function $y(t)$ has derivatives of all orders near t_0, then its value at any point t near t_0 can be determined from the values of its derivatives at t_0:

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- I.e., the value of the function at some point near t_0 is completely determined by its value at t_0 and the values of **all** of its derivatives at t_0
- Note that this works for time points in the past as well as those in the future, provided they are sufficiently close to t_0

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